PSG Institute of Technology and Applied Research

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2027_IV_CSBS_DBMS Lab

PSGITECH_2027_DBMS_Week 7_COD

Attempt : 1
Total Mark : 30
Marks Obtained : 30

Section 1 : Coding

1. Problem Statement

In a banking application, transactions related to fund transfers and balance updates are crucial for maintaining the integrity of customer accounts. To ensure the serializability of schedules and guarantee consistency in financial operations, the system must carefully manage multiple concurrent transactions.

The application involves two tables: bank_accounts and transaction_logs.

Sample Records are:

Two separate transactions need to be executed:

Transaction 1 (T1):

Deduct 100 from Account A's balance in the bank_accounts table.Roll back the deduction operation to restore the original balance of John's account. Insert the operation in transaction_logs.

Transaction 2 (T2):

Add 2000 to Account B's balance in the bank_accounts table.Commit the transaction to finalize the changes to Alice's account.Insert the operation in transaction_logs.Task:

Write the SQL queries for both transactions (T1 and T2) to achieve the above requirements.

Ensure the transactions are executed atomically using the START TRANSACTION statements.

Answer

file.sql

```
UPDATE bank_accounts
SET balance = balance - 100
WHERE account_id = 1;
INSERT INTO transaction_logs (account_id, transaction_type, amount)
VALUES (1, 'Debit', 100.00);
ROLLBACK;
START TRANSACTION;
UPDATE bank_accounts
SET balance = balance + 2000
WHERE account_id = 2;
INSERT INTO transaction_logs (transaction_id,
account_id,transaction_type,amount)
VALUES (2, 1, 'Credit', 2000.00);
COMMIT;
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

In a university's course registration system, two transactions, T1 and T2, are responsible for updating student course enrollments. Each transaction represents the enrollment of a student in different courses.

Table: student_enrollment

The following table has already been created, and the records are inserted into the table.

Tasks:

Write an SQL query to update the enrollment_status to 'Dropped' for the student with student_id = 1 and course_id = 102.Then, display all records in the student_enrollment table.Use the rollback command to undo the changes made in Task 1.Write an SQL query to update the enrollment_status to 'Enrolled' for the student with student_id = 2 and course_id = 101.After performing the update in Task 4, commit the changes to the database.

Answer

file.sql

```
UPDATE student_enrollment
SET enrollment_status = 'Dropped'
WHERE student_id = 1 AND course_id = 102;
SELECT * FROM student_enrollment;
UPDATE student_enrollment
SET enrollment_status = 'Enrolled'
WHERE student_id = 2 AND course_id = 101;
COMMIT;
UPDATE student_enrollment
SET enrollment_status = 'Enrolled'
WHERE student_id = 1 AND course_id = 102;
UPDATE student_enrollment
SET enrollment_status = 'Enrolled'
WHERE student_id = 2 AND course_id = 101;
COMMIT;
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

In an Inventory Management System, the database is used to track product inventory levels and product details. Two transactions, T1 and T2, are responsible for updating product information concurrently.

Table: product_inventory

Sample input records:

Design a system that ensures the consistency of product information while maintaining the serializability of transactions.

Transaction T1 (S1):

Update the quantity of "Product A" to 60.Increase the price of "Product A" by 5%.Commit the transaction.

Transaction T2 (S2):

Update the quantity of "Product B" to 90.Decrease the price of "Product B" by 10%.Retrieve all records from the product_inventory table to verify updates.Rollback the transaction, discarding all changes made to "Product B".

Finally, ensure that only the committed changes from T1 are reflected in the product_inventory table, while the updates from T2 are not applied due to the rollback.

Answer

<u>file.sql</u>

```
START TRANSACTION;
UPDATE product_inventory
SET quantity = 60
WHERE product_name = 'Product A';
UPDATE product_inventory
SET price = price * 1.05
WHERE product_name = 'Product A';
COMMIT;
START TRANSACTION;
```

```
UPDATE product_inventory
SET quantity = 90
WHERE product_name = 'Product B';
UPDATE product_inventory
SET price = price * 0.90
WHERE product_name = 'Product B';
SELECT * FROM product_inventory;
ROLLBACK;
```

Status : Correct ***Marks : 10/10***